

INTERNSHIP WORK LOG AND PROGRESS REPORT

Project Title:

Business Intelligence RAG Chatbot using Tesla Financial and Annual Report Data.

Introduction:

The objective of this internship project is to develop a Retrieval-Augmented Generation (RAG) based chatbot capable of answering business-related queries using Tesla's financial statements and annual reports. The chatbot will leverage Large Language Models (LLMs), vector databases, and retrieval techniques to provide context-aware and accurate responses.

Dataset Description

The project utilizes the following Tesla datasets:

- Financial Statement (2020-2023)
- Balance Sheet (2020-2023)
- Cash Flow Statement (2020-2023)
- Tesla Annual and Quarterly Reports

The structured datasets are used for KPI analysis and visualization, while the annual reports provide unstructured textual information for the RAG pipeline.

Data Preprocessing

- Renamed columns for consistency.
- Removed the 2019 column due to excessive missing values.
- Performed missing value analysis.
- Converted datasets from wide format to long format using `pd.melt()`.
- Created:
- `financial_long.csv`
- `balance_long.csv`
- `cashflow_long.csv`

Observations

- The datasets contain yearly financial information from 2020 to 2023.
- The 2019 records were sparse and unsuitable for analysis.
- Long-format datasets simplified time-series analysis and visualization.

Exploratory Data Analysis(EDA)

Visualizations Generated

- Total Revenue trend
- EBITIDA trend
- Net income trend
- Free Cash Flow trend

- Total Debt trend

Key Observations

- Revenue increased steadily from 2020 to 2023
- EBITDA peaked in 2022 and declined slightly in 2023
- Free Cash Flow showed strong growth until 2022, followed by a decline in 2023
- Total Debt reduced significantly until 2022 but increased again in 2023.

Key Performance Indicators(KPIs)

The following KPIs were identified:

1. Total Revenue
2. Net Income
3. EBITDA
4. Free Cash Flow
5. Total Debt

KPI Analysis

- Constructed KPI pivot tables
- Calculated Year over Year growth percentages
- Compared financial performance across years

Business Insights

Revenue Growth

Tesla's Total Revenue increased from approximately \$31.5 billion in 2020 to \$96.7 billion in 2023, demonstrating strong

business growth. The highest YoY growth was observed in 2021.

Profitability

Net Income and EBITDA increased substantially between 2020 and 2022, indicating improvements in profitability and operational efficiency.

Cash Flow

Free Cash Flow increased significantly until 2022. However, a decline was observed in 2023, suggesting higher capital expenditures or investments.

Debt Management

Total Debt decreased consistently from 2020 to 2022, reflecting improved financial stability. A rise in debt was observed in 2023 and will be investigated further.